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Multi-Granularity Autoformer for long-term deterministic and probabilistic power load forecasting

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ABSTRACT

Long-term power load forecasting is critical for power system planning but is constrained by intricate temporal patterns. Transformer-based models emphasize modeling long- and short-term dependencies yet encounter limitations from complexity and parameter overhead. This paper introduces a novel Multi-Granularity Autoformer (MG-Autoformer) for long-term load forecasting. The model leverages a Multi-Granularity Auto-Correlation Attention Mechanism (MG-ACAM) to effectively capture fine-grained and coarse-grained temporal dependencies, enabling accurate modeling of short-term fluctuations and long-term trends. To enhance efficiency, a shared query–key (Q–K) mechanism is utilized to identify key temporal patterns across multiple resolutions and reduce model complexity. To address uncertainty in power load forecasting, the model incorporates a quantile loss function, enabling probabilistic predictions while quantifying uncertainty. Extensive experiments on benchmark datasets from Portugal, Australia, America, and ISO New England demonstrate the superior performance of the proposed MG-Autoformer in long-term power load point and probabilistic forecasting tasks.

1. Introduction

Multi-horizon load forecasting, which involves forecasting power load at multiple future time steps, plays a vital role in modern power systems (Lim, Arık, Loeff, & Pfister, 2021; Yang, Zhou, Wu, Liu, & Wang, 2022). Unlike single-step forecast, this approach provides a detailed outlook across varying time horizons, enabling comprehensive decision-making for both short-term operations and long-term planning (Liu, Kong, Huang, Du, & Xiang, 2024; Smyl, Dudek, & Pełka, 2024). For instance, accurate multi-horizon forecasts support real-time load balancing, efficient energy storage management, and proactive grid maintenance (Liu & Du, 2023). Given its broad applications, improving the accuracy of multi-horizon load forecasting is crucial for improving grid resilience, optimizing resource allocation, and supporting sustainable energy development (Zhuang, Fan, Xia, & Zhu, 2023).

Deep learning and machine learning methods have been widely applied in power load forecasting due to their powerful capability to model complex temporal dependencies and nonlinear patterns (Wang, Wu, Cheng, Liu, & Qiu, 2024; Yang, Lou, Wu, Zhang and Gao, 2024). Recurrent Neural Networks (RNNs), such as Long Short-Term Memory Networks (LSTMs) (Abou Houran, Bukhari, Zafar, Mansoor, & Chen, 2023) and Gated Recurrent Units (GRUs) (Xiao, Zou, Chi and Fang, 2023), are widely employed for their ability to capture temporal dependencies in power load forecasting. Building on this foundation, advanced models like DeepAR (Salinas, Flunkert, Gasthaus, & Januschowski, 2020) further extend the utility of RNNs by effectively estimating the probability distribution of future time series. To improve feature extraction, Convolutional Neural Networks (CNNs) are widely integrated with RNNs. For instance, LSTNet (Lai, Chang, Yang, & Liu, 2018) combines CNNs with recursive jump connections to effectively capture both short-term and long-term temporal patterns. However, RNN-based methods, even with attention and convolutional enhancements, face limitations such as gradient vanishing and explosion, which hinder their ability to model long-term dependencies. This challenge becomes particularly significant in long-term forecasting, where prediction performance deteriorates as the forecast horizon extends (Zhou et al., 2021).

The self-attention-based Transformer framework has recently demonstrated exceptional performance in time sequence modeling (Vaswani et al., 2017; Wu, Xiao et al., 2020). Its ability to encode long-term time series without information loss makes it superior to RNNs for capturing long-range dependencies in load patterns (Kenton

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& Toutanova, 2019; Wan, Chang, AL-Bukhaiti, & He, 2023). The Transformer's encoder–decoder architecture efficiently extracts and processes features for forecast (Xiao, Liu, Fang, Liu and Wang, 2023). Several improved Transformer variants have been proposed to enhance its efficiency and accuracy. LogTrans (Li et al., 2019) integrates local convolution and logarithmic sparse attention to select time steps based on exponential intervals. Informer (Zhou et al., 2021) reduces computational complexity to $O(L \log L)$ using probability sparse attention based on KL-divergence. Autoformer (Wu, Xu, Wang, & Long, 2021) introduces an auto-correlation mechanism to extract periodicity and incorporates time series decomposition for gradual load series refinement. While Transformer-based models excel at capturing long-term dependencies, they face challenges in actual power load forecast. The quadratic complexity of self-attention with sequence length makes them computationally expensive, and efficiency improvements, such as sparse attention mechanisms, often reduce information utilization, limiting their ability to model intricate temporal patterns (Dosovitskiy et al., 2021; Huang et al., 2019).

Traditional load forecasting has focused on deterministic point forecasting, but the increasing integration of distributed renewable energy and smart grids has introduced greater randomness and uncertainty (Liu, Nowotarski, Hong, & Weron, 2015; Xie & Hong, 2016). Probabilistic forecasting addresses this challenge by quantifying prediction uncertainty (Rafiei, Niknam, Aghaei, Shafie-Khah, & Catalão, 2018). The main approaches include Interval Estimation (IE) (Guan, Luh, Michel, & Chi, 2013), Kernel Density Estimation (KDE) (Li, Yu, Huang, Sun, & Jia, 2023), and Quantile Regression (QR) (Zhang, Wang, Zhang, Wang, & Zhang, 2020). IE generates prediction intervals at preset confidence levels but lacks detailed probability distributions. KDE estimates the Probability Density Function (PDF) but relies on large sample sizes and proper bandwidth selection. QR, by predicting multiple quantiles, provides a flexible probability distribution without assuming a prior distribution, making it particularly effective for probabilistic load forecasting (Aprillia, Yang, & Huang, 2020; Bracale, Caramia, De Falco, & Hong, 2019; Chen, Kang, Chen, & Wang, 2020). These advancements highlight the need for accurate and robust uncertainty quantification, which this paper addresses using QR.

Despite these advancements, existing methods still face challenges in capturing long-term dependencies and effectively leveraging temporal patterns. RNN-based models suffer from gradient vanishing and explosion issues, limiting their ability to model extended temporal sequences. Transformer-based methods are effective for data with stable patterns; however, for power load data with significant distribution shifts and ambiguous patterns, accurately capturing periodicity requires handling much longer sequences, leading to elevated computational complexity. Additionally, the increasing randomness and uncertainty further complicate precise forecasting (Shao et al., 2024). In response, multi-scale feature fusion, which integrates information across different temporal resolutions, has gained traction in various applications such as signal processing (Wu, Jin, Daly, Wang and Cichocki, 2023), computer vision (Wu, Yao et al., 2023), and other tasks (Yang, Gao et al., 2024; Zhao et al., 2024) for its ability to extract comprehensive features across varying temporal scales, enabling richer representations and improved model performance.

To tackle these challenges, this paper proposes the Multi-Granularity Auto-Correlation Attention Mechanism (MG-ACAM). By integrating fine-grained and coarse-grained temporal features through multi-resolution fusion and series decomposition, MG-ACAM significantly improves both accuracy and robustness for long-term deterministic and probabilistic load forecasting. The main contributions of this work are as follows:

1. MG-ACAM is introduced to capture fine-grained and coarse-grained temporal dependencies effectively. It efficiently identifies key temporal patterns across multiple resolutions, enhancing the modeling of both short-term variations and long-term trends, and improving forecasting accuracy.
2. The architecture incorporates a series decomposition block to separate trend and seasonal components. Through multi-granularity analysis and the query–key (Q–K) mechanism, it captures longer periodicities while reducing computational complexity, ensuring robust performance in long-term forecasting tasks.
3. Extensive experiments on four benchmark datasets validate the superiority of MG-Autoformer. Notably, on the Portugal dataset at a 168-step deterministic forecast, it achieves a 17.2% reduction in MSE and a 9.0% reduction in MAE compared to Autoformer, demonstrating its enhanced accuracy and reliability in long-term forecasting tasks.

The paper is organized as follows: Section 2 outlines the problem formulation for long-term power load forecasting and details the theory and implementation of the proposed MG-Autoformer. Section 3 describes the datasets and evaluation metrics. Section 4 describes extensive experiments to assess both deterministic and probabilistic forecasting performance, along with ablation studies. Finally, Section 5 concludes the paper and discusses future research directions.

2. Multi-granularity autoformer for both deterministic and probabilistic forecasting

2.1. Overview

The objective of time series forecasting is to predict future values $\hat{\mathbf{Y}} \in \mathbb{R}^m$ based on historical observations $\mathbf{X} \in \mathbb{R}^{n \times l}$, where n denotes the number of variables, and l represents the input window size. Specifically, the historical observations are defined as $\mathbf{X} = \{\mathbf{x}_{t-l+1}, \mathbf{x}_{t-l+2}, \dots, \mathbf{x}_t\}$, where $\mathbf{x}_t \in \mathbb{R}^n$ contains the observations for all n variables at time step t. The forecasting process is mathematically expressed as:

$$\hat{\mathbf{Y}} = Q(\mathbf{X}), \tag{1}$$

where $\hat{\mathbf{Y}} = \{\hat{y}_{t+1}, \hat{y}_{t+2}, \dots, \hat{y}_{t+m}\}$ represents the predicted values for the next m time steps, and Q denotes the mapping function. The prediction horizon m varies depending on the application, spanning a few steps for short-term forecasting and extending to hundreds or more for long-term forecasting.

The forecasting task aims to minimize the discrepancy Δ_d between the predicted values $\hat{\mathbf{Y}}$ and the ground truth $\mathbf{Y}$. This optimization problem can be formulated as:

$$\Delta_d = \arg\min_{Q} \|\hat{\mathbf{Y}} - \mathbf{Y}\|, \tag{2}$$

where $\mathbf{Y} = \{y_{t+1}, y_{t+2}, \dots, y_{t+m}\}$ represents the observed value. A perfect prediction is achieved when $\Delta_d = 0$.

As highlighted in the introduction, time series forecasting faces challenges such as capturing long- and short-term dependencies and providing probabilistic predictions to quantify uncertainty. To address these issues, our proposed MG-Autoformer incorporates innovative mechanisms and architectural enhancements, as shown in Fig. 1:

1. **Multi-granularity auto-correlation attention mechanism**: Captures temporal dependencies across multiple resolutions by integrating fine-grained and coarse-grained features. The shared query–key (Q–K) efficiently identifies key temporal patterns, reducing model complexity.
2. **Encoder–decoder architecture with series decomposition:** Separates trend-cyclical and seasonal components for specialized processing. The encoder extracts multi-granularity features, and the decoder refines these features while initializing placeholders for future predictions.
3. **Quantile loss function for probabilistic forecasting:** Extends the model to probabilistic forecasting by using quantile loss, enabling uncertainty quantification and improving robustness for real-world applications.

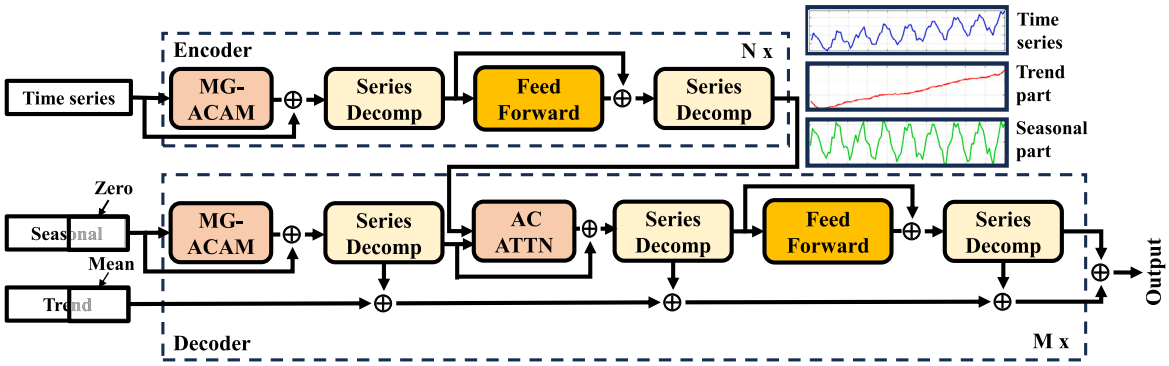


Fig. 1. The multi-granularity Autoformer.

2.2. Multi-granularity auto-correlation attention mechanism

The Multi-Granularity Auto-Correlation Attention Mechanism (MG-ACAM) is developed to capture temporal dependencies across multiple resolutions, facilitating the effective representation of both short-term and long-term patterns in time series data.

To extract multi-granularity representations, the input sequence $\mathbf{X} \in \mathbb{R}^{n\times l}$ is downsampled at different granularities $\{a_1, a_2, \ldots, a_m\}$. For each granularity a_i, the sequence length is reduced to $l' = l/a_i$ using a one-dimensional convolution operation defined as:

$$\mathbf{X}'_{c,j} = \sum_{k=1}^{n}\sum_{p=1}^{a_i} \mathbf{X}_{k,a_i\cdot j+p-1}\mathbf{W}_{c,k,p} + \mathbf{b}_c, \tag{3}$$

where $\mathbf{X}' \in \mathbb{R}^{d'\times l'}$ is the downsampled sequence, $\mathbf{W} \in \mathbb{R}^{d'\times n\times a_i}$ is the convolution kernel, $\mathbf{b} \in \mathbb{R}^{d'}$ is the bias term, c indexes the output channels, and j indexes the output time steps. By applying this operation across multiple granularities, the model generates a set of multi-granularity representations $\{\mathbf{X}'_{a_1}, \mathbf{X}'_{a_2}, \ldots, \mathbf{X}'_{a_m}\}$, which capture patterns at varying temporal resolutions. This multi-granularity approach allows the model to simultaneously focus on fine-grained details and broader trends.

For MG-ACAM, the downsampled sequence $\mathbf{X}'_{a_i} \in \mathbb{R}^{d'\times l'}$ is mapped into query ($\mathbf{Q}_{a_i}$), key ($\mathbf{K}_{a_i}$), and value ($\mathbf{V}_{a_i}$) matrices using linear transformations:

$$\mathbf{Q}_{a_i} = \mathbf{W}^Q_{a_i}\mathbf{X}'_{a_i}, \quad \mathbf{K}_{a_i} = \mathbf{W}^K_{a_i}\mathbf{X}'_{a_i}, \quad \mathbf{V}_{a_i} = \mathbf{W}^V_{a_i}\mathbf{X}'_{a_i}, \tag{4}$$

where $\mathbf{W}^Q_{a_i}, \mathbf{W}^K_{a_i}, \mathbf{W}^V_{a_i} \in \mathbb{R}^{d_q\times d'}$ are the mapping matrices. To simplify computations, the shared query–key (Q–K) mechanism aligns the query and key matrices by mapping them into the same feature space:

$$\mathbf{Q}_{a_i} = \mathbf{K}_{a_i} = \mathbf{W}^Q_{a_i}\mathbf{X}'_{a_i}, \quad \mathbf{V}_{a_i} = \mathbf{W}^V_{a_i}\mathbf{X}'_{a_i}. \tag{5}$$

The shared mechanisms significantly reduce the number of parameters and computational complexity while maintaining the ability to capture correlations effectively. In time series tasks, aligning query and key spaces enhances the computation of correlations across timestamps, improving temporal dependency modeling.

To further optimize attention for long-term dependencies, MG-ACAM incorporates the Auto-Correlation Function (ACF). The ACF identifies the most relevant lagged correlations, allowing the model to focus attention on key temporal relationships. The top c lags $\{\tau_1, \tau_2, \ldots, \tau_c\}$ with the highest correlation scores are selected as follows:

$$\{\tau_1, \tau_2, \ldots, \tau_c\} = \arg \operatorname*{Topc}_{\tau\in 1,\ldots,l'-1} \left(R_{\mathbf{Q}_{a_i},\mathbf{K}_{a_i}}(\tau)\right), \tag{6}$$

where the auto-correlation coefficient $R_{\mathbf{Q}_{a_i},\mathbf{K}_{a_i}}(\tau)$ is efficiently computed using Fast Fourier Transforms (FFT). The resulting correlation values are then normalized through the softmax function:

$$\hat{R}_{\mathbf{Q}_{a_i},\mathbf{K}_{a_i}}(\tau_1, \tau_2, \ldots, \tau_c) = \mathrm{SoftMax}\left(R_{\mathbf{Q}_{a_i},\mathbf{K}_{a_i}}(\tau_1), R_{\mathbf{Q}_{a_i},\mathbf{K}_{a_i}}(\tau_2), \ldots, R_{\mathbf{Q}_{a_i},\mathbf{K}_{a_i}}(\tau_c)\right). \tag{7}$$

Finally, the auto-correlation attention matrix is computed by applying the lag shifts to the value matrix $\mathbf{V}_{a_i}$ and weighting them with the normalized correlations:

$$\text{Auto-Correlation}(\mathbf{Q}_{a_i}, \mathbf{K}_{a_i}, \mathbf{V}_{a_i}) = \sum_{i=1}^{c} \mathrm{Roll}\left(\mathbf{V}_{a_i}, \tau_i\right) \cdot \hat{R}_{\mathbf{Q}_{a_i},\mathbf{K}_{a_i}}(\tau_i), \tag{8}$$

where $\mathrm{Roll}(\mathbf{V}_{a_i}, \tau_i)$ represents a circular shift operation that displaces the elements of the sequence $\mathbf{V}_{a_i}$ by τ_i positions along the time axis. Elements shifted beyond the first position are reintroduced at the last position, maintaining the sequence's continuity. Fig. 2 illustrates this circular shift mechanism.

By integrating multi-granularity representations, a shared Q–K mechanism, and auto-correlation attention, MG-ACAM, as illustrated in Fig. 3, significantly improves temporal feature extraction while boosting computational efficiency.

2.3. Encoder–decoder architecture with series decomposition

This paper leverages the Transformer architecture for time series forecasting. The overall framework named MG-Autoformer, as shown in Fig. 1, consists of an encoder–decoder structure combined with a series decomposition block and MG-ACAM for efficient modeling of temporal patterns.

The series decomposition block separates the input time series into two components: the trend-cyclical part and the seasonal part. The trend-cyclical part captures long-term progression, while the seasonal part models periodic variations. The decomposition is defined as:

$$\mathbf{X}_t = \mathrm{AvgPool}(\mathrm{Padding}(\mathbf{X})), \tag{9}$$

$$\mathbf{X}_s = \mathbf{X} - \mathbf{X}_t, \tag{10}$$

where $\mathbf{X} \in \mathbb{R}^{n\times l}$ represents the input time series of length l with n dimensions, and $\mathbf{X}_t$ and $\mathbf{X}_s$ denote the trend-cyclical and seasonal components, respectively. This block is embedded within both encoder and decoder layers for progressive refinement.

The encoder models the seasonal patterns from the historical input $\mathbf{X}_{\text{en}} \in \mathbb{R}^{n\times l}$. It consists of stacked layers that perform the following operations:

$$\mathbf{S}^d_{\text{en}} = \mathrm{SeriesDecomp}(\text{MG-ATTN}(\mathbf{X}^{d-1}_{\text{en}}) + \mathbf{X}^{d-1}_{\text{en}}), \tag{11}$$

$$\mathbf{X}^d_{\text{en}} = \mathrm{FeedForward}(\mathbf{S}^d_{\text{en}}) + \mathbf{S}^d_{\text{en}}, \tag{12}$$

where $\mathbf{S}^d_{\text{en}}$ represents the seasonal component at layer d. The feed-forward module is a two-layer fully connected network with a non-linear activation. The final encoder output $\mathbf{X}^N_{\text{en}}$ provides refined seasonal information for the decoder.

The decoder predicts the future time series by modeling both trend and seasonal components. Its input is structured as:

$$\mathbf{X}_{\text{de},s} = \mathrm{Concat}(\mathbf{X}_{\text{en},s}, \mathbf{X}_0), \tag{13}$$

$$\mathbf{X}_{\text{de},t} = \mathrm{Concat}(\mathbf{X}_{\text{en},t}, \mathbf{X}_{\text{Mean}}), \tag{14}$$

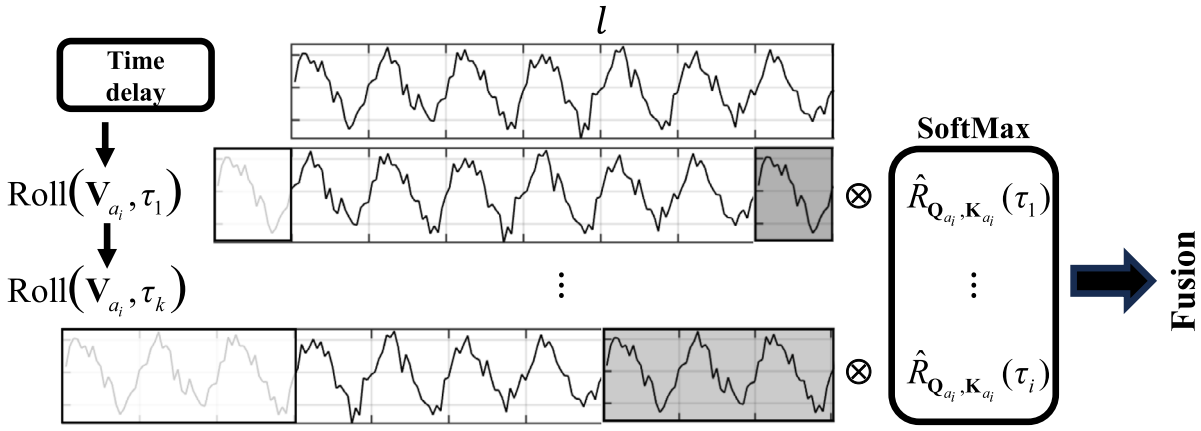


Fig. 2. The illustration of the ROLL circular shift operation.

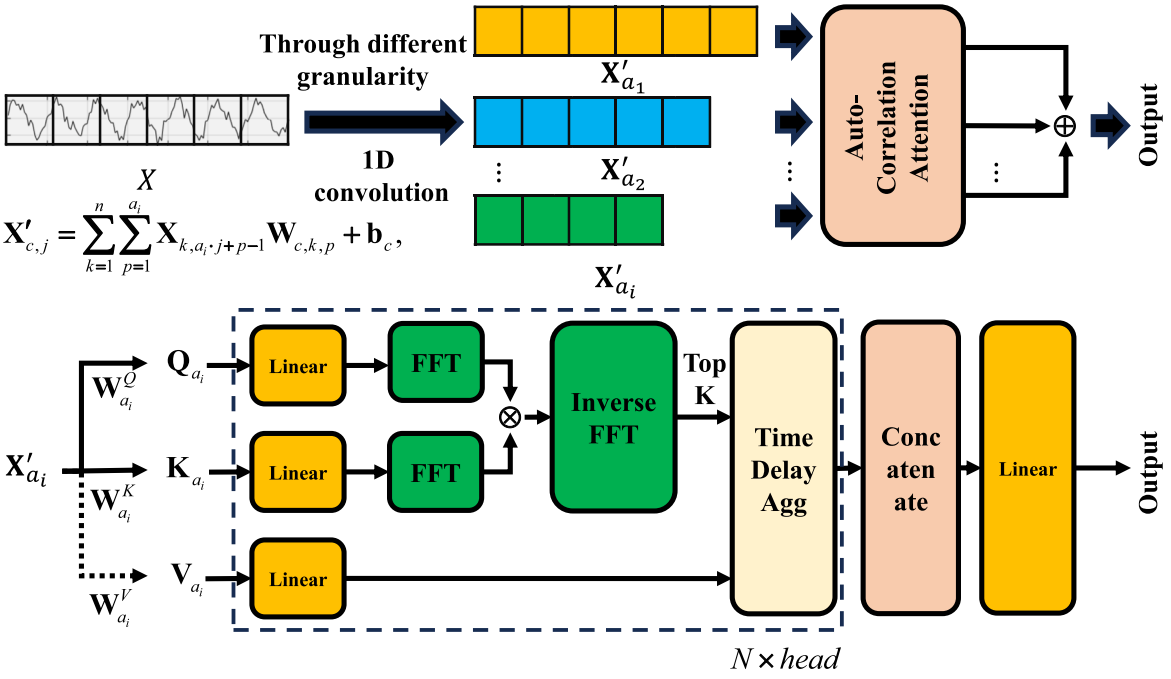


Fig. 3. The multi-granularity auto-correlation attention mechanism.

where $\mathbf{X}_0, \mathbf{X}_{\text{Mean}} \in \mathbb{R}^{n\times m}$ are placeholders for the seasonal and trend components, respectively, with m as the prediction horizon. The seasonal input $\mathbf{X}_{\text{de},s} \in \mathbb{R}^{n\times(l/2+m)}$ consists of the latter half of the encoder's seasonal component, $\mathbf{X}_{\text{en},s} \in \mathbb{R}^{n\times l/2}$, concatenated with zero-filled placeholders $\mathbf{X}_0$. The trend input $\mathbf{X}_{\text{de},t} \in \mathbb{R}^{n\times(l/2+m)}$ combines the encoder's trend component, $\mathbf{X}_{\text{en},t} \in \mathbb{R}^{n\times l/2}$, with mean-filled placeholders $\mathbf{X}_{\text{Mean}}$, where $\mathbf{X}_{\text{Mean}}$ contains the mean value of the historical input $\mathbf{X}_{\text{en}}$ along the time dimension. This setup provides the decoder with refined historical features and initialized placeholders for future predictions.

Each decoder layer performs the following operations:

$$\mathbf{S}^d_{\text{de}}, \mathbf{T}^d_{\text{de}} = \text{SeriesDecomp}(\text{MG-ATTN}(\mathbf{X}^{d-1}_{\text{de}}) + \mathbf{X}^{d-1}_{\text{de}}), \tag{15}$$

$$\mathbf{X}^d_{\text{de}} = \text{FeedForward}(\mathbf{S}^d_{\text{de}}) + \mathbf{S}^d_{\text{de}}. \tag{16}$$

The seasonal component $\mathbf{S}^d_{\text{de}}$ is progressively refined through the decoder layers, with the output of each layer contributing to more accurate modeling of periodic variations. Together, the refined seasonal component $\mathbf{S}^d_{\text{de}}$ and accumulated trend component $\mathbf{T}^d_{\text{de}}$ are combined to form the final prediction.

2.4. Quantile forecast

The MG-Autoformer extends from a deterministic point forecast model to support probabilistic forecasting. For point forecasting, the loss function is Mean Squared Error (MSE). For probabilistic forecasting, quantile loss is used to evaluate performance across multiple quantiles and forecast time steps. Prediction intervals are generated by forecasting various percentiles simultaneously. The quantile loss is defined as:

$$L = \frac{1}{|Q| \cdot |T|} \sum_{q\in Q} \sum_{t\in T} QL_{q,t}(y_t, \hat{y}_t), \tag{17}$$

where $QL_{q,t}(y_t, \hat{y}_t)$ is the quantile loss for quantile q at time step t, $|Q|$ is the number of quantiles, and $|T|$ is the number of time steps. Quantile forecasts are generated through a linear transformation of the decoder output:

$$\hat{y}_{q,t} = W_q \tilde{\psi}(t, \tau) + b_q, \tag{18}$$

where $W_q \in \mathbb{R}^{1\times d}$ and $b_q \in \mathbb{R}$ are linear coefficients for quantile q, and $\tilde{\psi}(t, \tau)$ is the temporal fusion decoder output. The quantile loss

Table 1
Power load datasets description.

Dataset	Description	Time range	Granularity
Portuga	6 substation load series from Portugal, part of 'LD2011–2014.txt' published in UCI	Jan 1, 2013–Dec 31, 2014	1 h
Australia	Load series from 5 Australian states (NSW, QLD, SA, TAS, VIC)	Jan 1, 2019–Dec 31, 2020	1 h
America	Power load data from 18 U.S. regions, part of GEFCOM2012	Jan 1, 2004–Dec 31, 2005	1 h
New England	Load series of ISO New England control area	Jan 1, 2020–Dec 31, 2021	1 h

$QL_{q,t}(y_t, \hat{y}_t)$ is given by:

$$QL_{q,t}(y_t, \hat{y}_t) = q(y_t - \hat{y}_t)_+ + (1-q)(\hat{y}_t - y_t)_+, \tag{19}$$

where $(\cdot)_+ = \max(0, \cdot)$. The quantiles $Q = \{0.1, 0.5, 0.9\}$ are used, generating forecasts only for future horizons $\tau \in \{1, \ldots, m\}$. This approach effectively integrates point and probabilistic forecasts.

2.5. Complexity analysis of MG-ACAM

In the auto-correlation attention mechanism of our MG-Autoformer, the most correlated lags identified from the ACF are selected to compute correlation weights. To reduce computational complexity, FFT is applied based on the Wiener–Khinchin theorem (Wiener, 1930), resulting in a computational complexity of $O(d\,L \log L)$.

For an auto-correlation attention mechanism with granularity a_i, where Q, K and V are mapped to sequences of length L/a_i, the computational complexity is $O(d(L/a_i)\log(L/a_i))$. With multiple attention branches at different granularities, the overall computational complexity of MG-ACAM is:

$$O\left(d(L/a_1)\log(L/a_1) + \cdots + d(L/a_m)\log(L/a_m)\right). \tag{20}$$

Compared to single-granularity attention with a complexity of $O(d\,L \log L)$, the multi-granularity design of MG-ACAM reduces sequence lengths to L/a_i for each branch, balancing computational cost and model performance. The selected granularities $a_1 = 1$, $a_2 = 8$, and $a_3 = 12$ effectively capture both fine-grained and periodic features from hourly data, enhancing prediction accuracy while maintaining computational efficiency.

3. Dataset and performance indices

3.1. Dataset description

This paper uses four datasets for multi-horizon power load forecasting. The details of the dataset are shown in Table 1.

Based on the autocorrelation patterns shown in Fig. 4, granularities of 1, 8, and 12 are selected to effectively capture dependencies at different scales. A granularity of 1 focuses on short-term dependencies with high autocorrelation at lower lags, granularity 8 captures moderate periodicity observed in datasets like Australia and New England, and granularity 12 reflects long-term seasonal patterns prominent in datasets such as America and New England.

3.2. Evaluation criteria

This paper employs four evaluation metrics to assess forecast performance: Mean Absolute Error (MAE), Mean Squared Error (MSE), Weighted Score (WS), and Quantile Function (QF). MAE measures the average magnitude of errors, while MSE emphasizes large errors through squared terms. WS evaluates prediction intervals by considering both width and penalties for confidence level violations. QF assesses prediction intervals across multiple quantiles.

Table 2 summarizes the metrics for point and probabilistic prediction evaluation. Here, T is the total number of samples, y_t and $\hat{y}_t$ are the observed and predicted values, L_t and U_t denote the lower and upper bounds of the prediction interval, $\alpha = 0.1$ is the significance level, and $\delta_t = U_t - L_t$ represents the interval width, and Ω denotes the set of all samples within the prediction interval.

Table 2
Evaluation metrics for point and probabilistic forecast.

Metric	Formula
Mean Absolute Error (MAE)	$\text{MAE} = \frac{1}{T}\sum_{t=1}^{T}\lvert y_t - \hat{y}_t\rvert$
Mean Squared Error (MSE)	$\text{MSE} = \frac{1}{T}\sum_{t=1}^{T}(y_t - \hat{y}_t)^2$
Winkler Score (WS)	$\text{WS} = \sum_{y_t \in \Omega}\delta_t + \begin{cases} 0, & U_t \geq y_t \geq L_t \\ 2(L_t - y_t)/\alpha, & L_t \geq y_t \\ 2(y_t - U_t)/\alpha, & y_t \geq U_t \end{cases}$ where $\delta_t = U_t - L_t$
Quantile Function (QF)	$L = \frac{1}{\lvert Q\rvert \cdot \lvert T\rvert}\sum_{q\in Q}\sum_{t\in T} QL_{q,t}(y_t, \hat{y}_t)$,

4. Case study

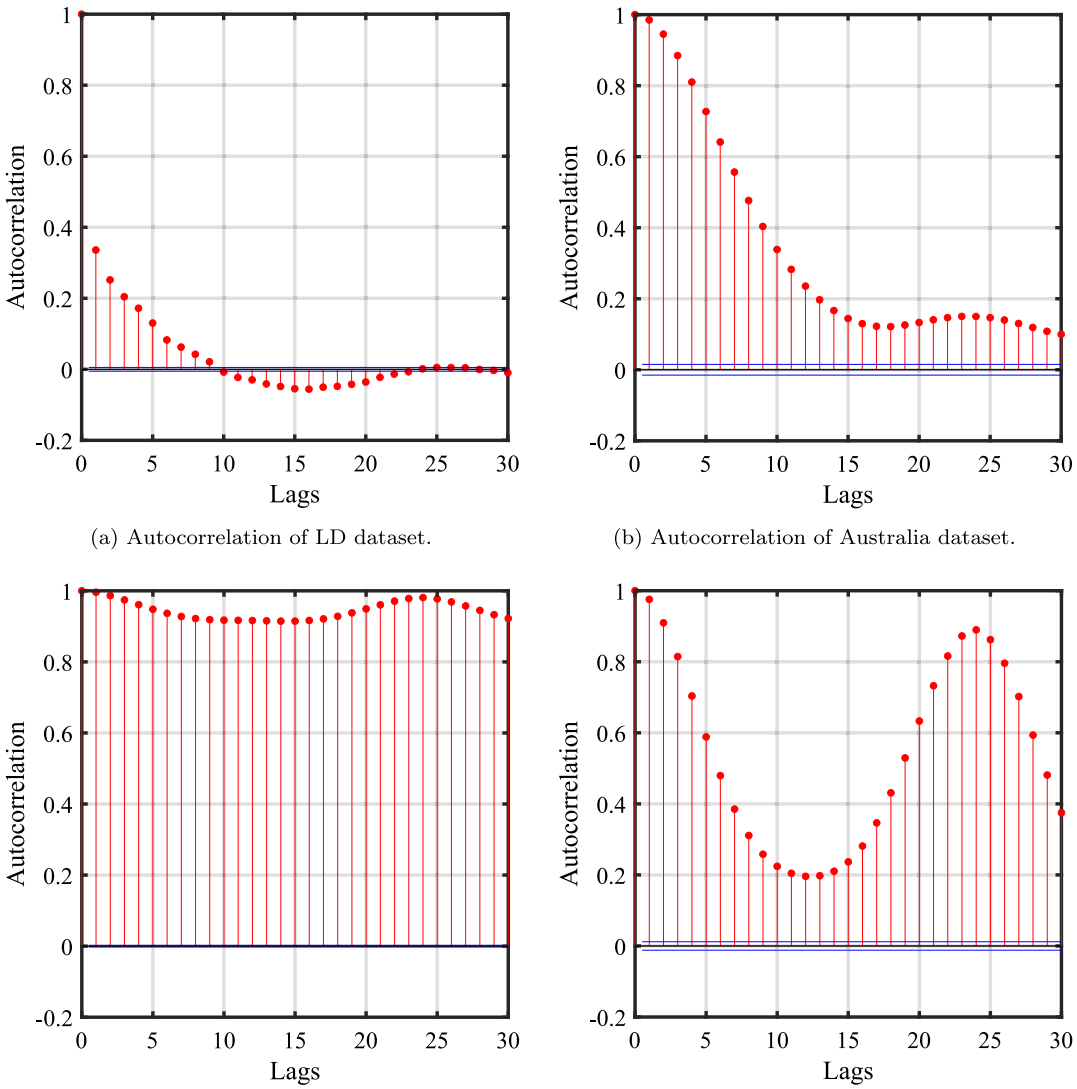
4.1. Experimental setup

In the deterministic point prediction experiment. The comparison models for long-term forecasting include Autoformer (Wu et al., 2021), Informer (Zhou et al., 2021), Transformer (Vaswani et al., 2017), and MTGNN (Wu, Pan et al., 2020). In the probabilistic forecasting experiment, DeepAR (Salinas et al., 2020), DeepTCN (Chen et al., 2020) and TFT (Lim et al., 2021) are included as probabilistic forecasting comparison models. The hyperparameter settings are shown in Table 3.

The dataset is split into training, validation, and test sets with a ratio of 6:2:2. The learning rate is set to 0.0001, and the dropout rate is set to 0.2. Training typically stops within 10 epochs if the loss of the validation set does not decline. All experiments are implemented using PyTorch and executed on an NVIDIA GTX 1650TI. For forecasting steps of 48, 168, and 360, the input sequence length is fixed at 96. Batch sizes are set to 32 for 48-step and 168-step predictions, and 16 for 360-step predictions.

4.2. Deterministic point forecasting performance

MG-Autoformer exhibits significant improvements in deterministic forecasting across all datasets and forecast horizons, as detailed in Table 4. By leveraging the MG-ACAM, the model effectively captures temporal dependencies at varying resolutions, leading to enhanced accuracy for both short- and long-term forecasts. For example, on the Portugal dataset, MG-Autoformer achieves a 17.2% reduction in MSE (0.303 → 0.251) and a 9.0% reduction in MAE (0.376 → 0.342) for the 168-step forecast. In the 360-step forecast, the model further reduces MSE by 9.1% (0.330 → 0.300) and MAE by 5.6% (0.408 → 0.385) compared to Autoformer. These results demonstrate the effectiveness of MG-ACAM in modeling medium- and long-term dependencies, where traditional models often struggle to maintain accuracy. Similar performance gains are observed on other datasets, such as Australia and New England, confirming the generalizability of the approach.



(a) Autocorrelation of LD dataset.

(b) Autocorrelation of Australia dataset.

(c) Autocorrelation of America dataset.

(d) Autocorrelation of New England dataset.

Fig. 4. Comparison of autocorrelation in different datasets.

Table 3
Hyperparameter settings for comparison models and the proposed method.

Model	Hyperparameter settings
MG-Autoformer	Encoder layers = 3; Decoder layers = 2; Heads = 4; Multi-granularity blocks = 3; c = 3
Autoformer	Encoder layers = 3; Decoder layers = 2; Heads = 4; c=3
Informer	Encoder layers = 3; Decoder layers = 2; Heads = 4
Transformer	Encoder layers = 3; Decoder layers = 2; Heads = 8
MTGNN	Channels = 16; K = 2
DeepAR	RNN layers = 2; RNN units = 32
DeepTCN	Filters = 128; Kernel Size = 5
TFT	Encoder layers = 2; Decoder layers = 1; Heads = 2; Hidden layer size = 128

Fig. 5 illustrates the deterministic point forecasting performance of MG-Autoformer on the Portugal and America datasets for 48-step, 168-step, and 360-step horizons. The model closely aligns with the observed data in short-term (48-step) and medium-term (168-step) forecasts, effectively capturing periodic patterns and trends. For long-term forecasts (360-step), MG-Autoformer remains robust, maintaining alignment with the real data on the Portugal dataset while showing slight deviations in the America dataset due to distribution shifts. These deviations highlight the model's sensitivity to rare patterns in extended horizons, a challenge inherent to long-sequence forecasting tasks. Nevertheless, the results validate the ability of MG-Autoformer to model short- and long-term dependencies with high precision, leveraging its multi-granularity design.

4.3. Probabilistic forecasting performance

In probabilistic forecasting, MG-Autoformer demonstrates superior performance in quantifying uncertainty, as shown in Table 5. Across most datasets and forecast horizons, the model achieves the lowest WS and QF criteria, showcasing its ability to generate reliable prediction intervals. For instance, on the New England dataset's 360-step forecast, MG-Autoformer reduces WS by 9.1% (1326→ 1205) and QF by 11.8%

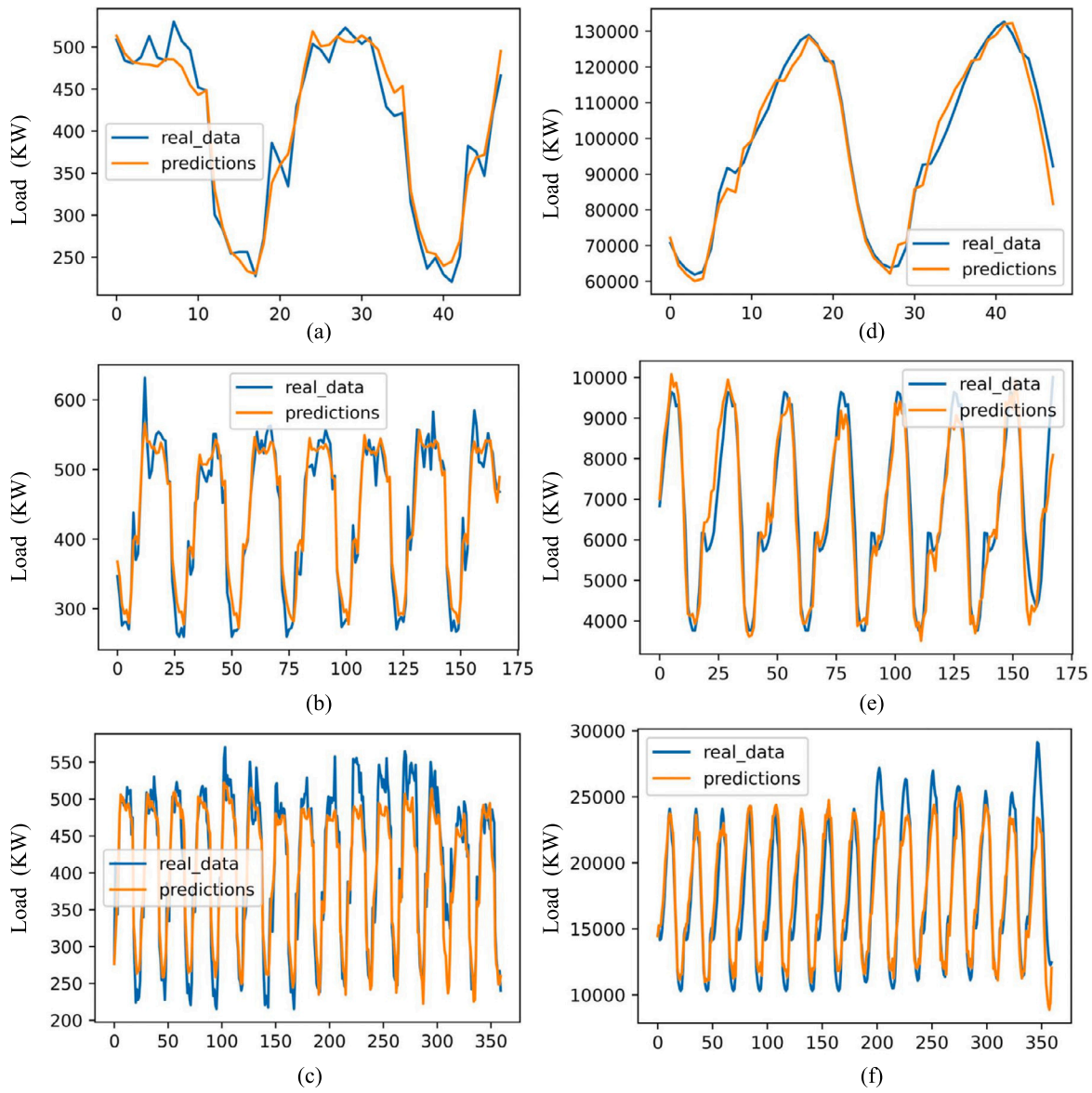


Fig. 5. Deterministic point forecasting results: 48-step, 168-step, and 360-step forecasts for Portugal dataset (a–c) and America dataset (d–f).

Table 4
Deterministic point forecasting comparison of different models at various forecast steps.

Dataset	Forecast step	MG-Autoformer		Autoformer		Informer		Transformer		MTGNN	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Portugal	48	**0.194**	**0.304**	0.226	0.333	0.257	0.343	0.229	0.329	0.225	0.323
	168	**0.251**	**0.342**	0.303	0.376	0.374	0.404	0.425	0.428	0.304	0.375
	360	**0.300**	**0.385**	0.330	0.408	0.380	0.432	0.385	0.426	0.366	0.426
Australia	48	**0.263**	**0.375**	0.352	0.446	0.288	0.380	0.284	0.382	0.375	0.474
	168	**0.398**	**0.451**	0.418	0.481	0.432	0.452	0.463	0.466	0.399	0.484
	360	**0.375**	**0.444**	0.360	0.443	0.417	0.464	0.397	0.453	0.463	0.529
America	48	**0.318**	**0.412**	0.377	0.466	0.357	0.441	0.402	0.456	0.400	0.458
	168	**0.504**	**0.535**	0.525	0.555	0.578	0.573	0.529	0.549	0.577	0.574
	360	0.599	0.622	**0.571**	**0.569**	0.621	0.594	0.557	0.572	0.636	0.610
New England	48	**0.331**	**0.420**	0.593	0.581	0.535	0.476	0.500	0.490	0.425	0.450
	168	**0.459**	**0.492**	0.642	0.552	0.567	0.510	0.697	0.577	0.518	0.523
	360	**0.385**	**0.458**	0.490	0.527	0.589	0.547	0.676	0.566	0.520	0.539

Table 5
Probabilistic forecasting comparison of different models at various forecast steps.

Dataset	Forecast step	MG-Autoformer		Autoformer		Informer		DeepAR		DeepTCN		TFT	
		WS	QF	WS	QF	WS	QF	WS	QF	WS	QF	WS	QF
Portugal	48	718	0.175	937	0.243	**642**	**0.153**	839	0.200	930	0.232	3754	0.279
	168	**2752**	**0.202**	3107	0.222	3760	0.228	4030	0.208	3219	0.236	5207	0.354
	360	**7035**	0.232	10 106	0.338	7122	**0.215**	8462	0.226	7409	0.246	7821	0.500
Australia	48	**617**	**0.180**	888	0.244	760	0.196	666	0.207	720	0.220	758	0.227
	168	**2400**	**0.207**	2544	0.220	3265	0.241	3258	0.271	2652	0.222	2884	0.223
	360	**5346**	**0.210**	5670	0.221	6527	0.229	7217	0.255	6396	0.239	8451	0.278
America	48	2748	**0.217**	3895	0.301	**2599**	0.229	3996	0.276	41 113	0.297	3359	0.312
	168	**11 087**	**0.261**	11 275	0.269	13 260	0.271	18 951	0.301	18 861	0.373	16 583	0.382
	360	**27 242**	**0.279**	32 813	0.320	33 136	0.350	39 278	0.358	36 642	0.340	43 017	0.443
New England	48	**140**	**0.214**	149	0.275	211	0.258	136	0.228	184	0.255	676	0.261
	168	**595**	**0.242**	687	0.271	872	0.347	644	0.250	711	0.277	854	0.298
	360	**1205**	**0.232**	1326	0.263	2074	0.364	1558	0.277	1464	0.273	1394	0.256

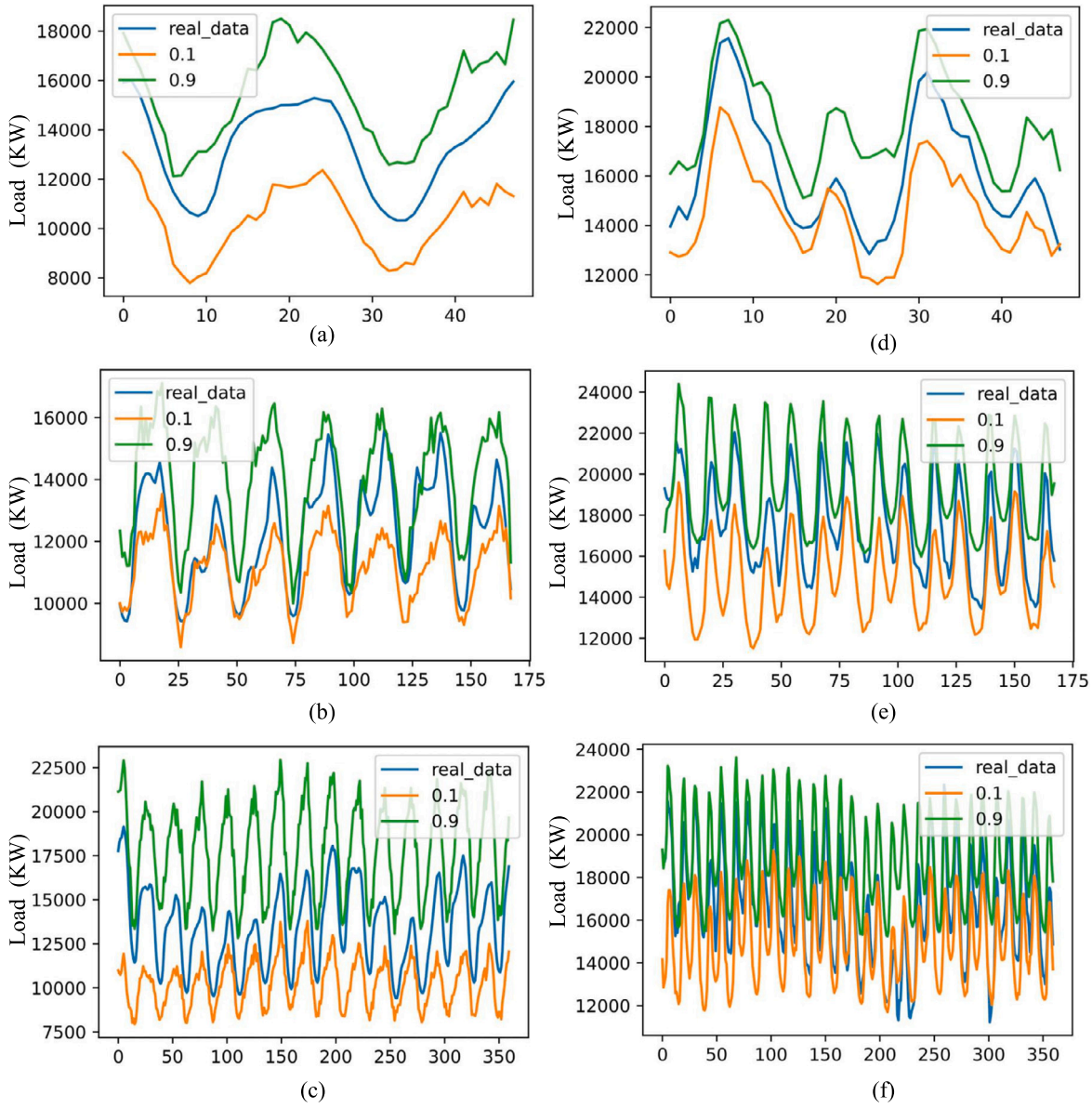


Fig. 6. Probabilistic forecasting results: 48-step, 168-step, and 360-step forecasts for New England dataset (a–c) and Australia dataset (d–f).

(0.263→0.232) compared to Autoformer, highlighting its robustness in long-term probabilistic forecasting. Additionally, on the Portugal dataset, MG-Autoformer improves WS by 30.4% (10106→7035) and QF by 31.4% (0.338→0.232) in the 360-step forecast, demonstrating its ability to maintain consistent performance across different datasets.

Fig. 6 visualizes the probabilistic forecasting results for the New England and Australia datasets. The 0.1 and 0.9 quantile prediction intervals effectively encompass the majority of the observed data, even in datasets with complex variability. This illustrates MG-Autoformer's capability to generate accurate and reliable prediction intervals,

Table 6
Ablation study on shared Q–K mechanism for deterministic point prediction.

Forecast step	Model	Portugal		Australia		America		New England	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
48	Autoformer (w/o Q–K)	0.226	0.333	0.352	0.446	0.377	0.466	0.593	0.581
	Autoformer	0.221	0.328	0.301	0.404	0.429	0.519	0.520	0.544
	MG-Autoformer (w/o Q–K)	0.199	0.308	**0.263**	0.377	0.339	0.429	0.441	0.483
	MG-Autoformer	**0.194**	**0.304**	**0.263**	**0.375**	**0.318**	**0.412**	**0.331**	**0.420**
168	Autoformer (w/o Q–K)	0.303	0.376	0.418	0.481	0.525	0.555	0.642	0.552
	Autoformer	0.267	0.349	**0.381**	0.460	**0.443**	**0.519**	0.535	0.533
	Informer (w/o Q–K)	0.363	0.424	0.544	0.569	0.580	0.596	0.623	0.611
	Informer	0.357	0.414	0.532	0.561	0.597	0.604	0.599	0.591
	Transformer (w/o Q–K)	0.310	0.387	0.410	0.469	0.526	0.564	0.775	0.645
	Transformer	0.276	0.370	0.396	0.465	0.473	0.531	0.840	0.681
	MG-Autoformer (w/o Q–K)	0.292	0.371	0.418	0.480	0.519	0.551	0.646	0.558
	MG-Autoformer	**0.251**	**0.342**	0.398	**0.451**	0.504	0.535	**0.459**	**0.492**
360	Autoformer (w/o Q–K)	0.330	0.408	0.360	0.443	0.571	**0.569**	0.490	0.527
	Autoformer	0.302	0.392	0.402	0.476	0.555	0.589	0.459	0.512
	MG-Autoformer (w/o Q–K)	0.304	**0.384**	**0.354**	**0.440**	**0.528**	0.586	0.468	0.509
	MG-Autoformer	**0.300**	0.385	0.375	0.444	0.599	0.622	**0.385**	**0.458**

Table 7
Ablation study on shared Q–K mechanism and MG-ACAM for probabilistic forecasting.

Forecast step	Model	Portugal		Australia		America		New England	
		WS	QF	WS	QF	WS	QF	WS	QF
48	Autoformer (w/o Q–K)	937	0.242	888	0.244	3895	0.301	149	0.275
	Autoformer	1020	0.240	802	0.229	3631	0.278	144	0.274
	MG-Autoformer (w/o Q–K)	**717**	**0.175**	638	**0.179**	2798	**0.229**	**125**	0.216
	MG-Autoformer	718	**0.175**	**617**	0.180	**2748**	**0.229**	140	**0.214**
168	Autoformer (w/o Q–K)	3107	0.222	2544	0.220	11 275	0.269	687	0.271
	Autoformer	3291	0.227	2504	0.219	13 518	0.295	**540**	**0.224**
	MG-Autoformer (w/o Q–K)	3445	0.232	**2304**	**0.203**	11 431	0.264	642	0.252
	MG-Autoformer	**2752**	**0.202**	2400	0.207	**11 087**	**0.261**	595	0.242
360	Autoformer (w/o Q–K)	10 106	0.338	5670	0.221	32 813	0.320	1326	0.263
	Autoformer	8522	0.277	5639	0.221	28 714	0.292	1529	0.273
	Informer (w/o Q–K)	7625	0.272	6102	0.265	26 555	0.296	1679	0.330
	Informer	7734	0.276	6206	0.269	27 472	0.303	1589	0.324
	Transformer (w/o Q–K)	7163	0.225	4996	0.200	25 663	0.267	1551	0.285
	Transformer	6917	0.229	**4812**	**0.198**	**25 300**	**0.266**	1972	0.325
	MG-Autoformer (w/o Q–K)	**6887**	**0.223**	5524	0.213	30 355	0.303	1440	0.257
	MG-Autoformer	7053	0.232	5346	0.210	27 242	0.279	**1205**	**0.232**

particularly for scenarios with significant uncertainty. By leveraging MG-ACAM, the model captures both fine-grained and coarse-grained temporal patterns, ensuring robust performance across diverse datasets.

4.4. Ablation study on shared Q-K mechanism

The ablation studies comprehensively evaluate the impact of the shared Q–K mechanism and MG-ACAM on MG-Autoformer's forecasting performance, with additional comparisons to Transformer and Informer. Tables 6 and 7 indicate that the shared Q–K mechanism reduces model complexity yet slightly increases prediction errors. For instance, in the 360-step deterministic forecast on the Portugal dataset, MSE rises by 2.6% (0.384→0.385). Probabilistic forecasting also shows a 2.4% increase in WS (6887→7053) and a 3.9% increase in QF (0.223→0.232), suggesting limited benefits in capturing temporal dependencies.

Meanwhile, MG-ACAM generally enhances long-term modeling, though its effectiveness varies. In deterministic forecasting for Portugal at 168 steps, MSE decreases by 6.0% (0.267→0.251), indicating improved handling of extended horizons. However, probabilistic results reveal occasional variability; for example, WS grows by 9.2% (540→595) in New England at 168 steps, implying that MG-ACAM's contributions depend on dataset characteristics and forecasting tasks.

4.5. Performance analysis and discussion

The experimental results demonstrate that MG-Autoformer consistently outperforms baseline models, particularly on datasets with clear seasonal patterns such as Portugal, Australia, and New England. These improvements highlight the MG-ACAM's effectiveness in capturing fine-grained variations and long-term periodic dependencies. The shared query–key mechanism further enhances cross-scale feature alignment, contributing to more stable and accurate long-term predictions. It is important to note that these improvements result from the coordinated integration of multiple architectural enhancements, rather than from the use of individual techniques such as quantile loss or multi-level attention, which are already common in time series forecasting research. MG-Autoformer's strength lies in its ability to fuse multi-scale features and maintain consistent attention across granularities, which is particularly valuable in long-term forecasting.

At the same time, the results also highlight that MG-Autoformer's advantages are dataset-dependent. On datasets with weak periodicity or irregular patterns, the performance gap between MG-Autoformer and other models is smaller. This reflects a broader understanding in time series forecasting — no single model excels across all datasets and forecasting scenarios, as model suitability depends heavily on dataset characteristics.

Finally, MG-Autoformer follows a progressive enhancement approach, building on the successful foundation of Autoformer while introducing targeted innovations to overcome its limitations. This approach, balancing inheritance and innovation, is consistent with the evolution of other Transformer-based forecasting models. The combination of these enhancements enables MG-Autoformer to better capture multi-scale dependencies and long-term patterns, making it particularly effective for long-horizon forecasting tasks.

Table A.8
Comparison results of long-term point predictions for the multi-granularity Autoformer.

	Step	MG-Autoformer		Autoformer		Informer		Reformer		LSTNet	
		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Electricity	96	**0.190**	**0.306**	0.201	0.317	0.274	0.368	0.312	0.402	0.680	0.645
	192	**0.204**	**0.317**	0.222	0.334	0.296	0.386	0.348	0.433	0.725	0.676
	336	**0.212**	**0.325**	0.231	0.338	0.300	0.394	0.450	0.433	0.828	0.727
Exchange	96	**0.178**	**0.317**	0.197	0.323	0.847	0.752	1.065	0.829	1.551	1.058
	192	**0.283**	0.390	0.300	**0.369**	1.204	0.895	1.188	0.906	1.477	1.028
	336	**0.442**	**0.490**	0.509	0.524	1.672	1.036	1.357	0.976	1.507	1.031
Traffic	96	0.648	0.391	**0.613**	**0.388**	0.719	0.391	0.732	0.423	1.107	0.685
	192	0.626	0.391	**0.616**	**0.382**	0.696	0.379	0.733	0.420	1.157	0.706
	336	0.643	0.410	**0.622**	**0.337**	0.777	0.420	0.742	0.420	1.216	0.730
Weather	96	**0.214**	**0.292**	0.266	0.336	0.300	0.384	0.689	0.596	0.594	0.587
	192	**0.281**	**0.350**	0.307	0.367	0.598	0.544	0.752	0.638	0.560	0.565
	336	**0.329**	**0.373**	0.359	0.395	0.578	0.523	0.639	0.596	0.597	0.587
ILI	24	**3.146**	**1.221**	3.483	1.287	5.764	1.677	4.400	1.382	6.026	1.770
	36	**2.959**	1.161	3.103	**1.148**	4.755	1.467	4.783	1.448	5.340	1.668
	60	2.995	1.154	**2.770**	**1.125**	5.264	1.564	4.882	1.483	5.548	1.720
ETT	48	**0.309**	0.377	0.312	**0.373**	1.457	1.001	1.871	1.735	3.567	1.687
	168	**0.415**	**0.428**	0.457	0.455	3.489	1.515	4.660	1.846	3.242	2.513
	336	**0.463**	**0.470**	0.471	0.475	2.723	1.340	4.028	1.668	2.544	2.591

5. Conclusion

Long-term power load forecasting is crucial yet challenging due to complex temporal patterns and uncertainty. This paper introduces MG-Autoformer, which integrates the MG-ACAM and a shared Q–K mechanism to effectively model both fine-grained and coarse-grained temporal dependencies while reducing computational complexity. The model also incorporates a quantile loss function, enabling seamless transitions between deterministic and probabilistic forecasting. Extensive experiments on benchmark datasets from Portugal, Australia, America, and ISO New England demonstrate that MG-Autoformer outperforms competing models, including Autoformer, Informer, and Transformer, in both accuracy and efficiency for long-term forecasting tasks. Future work will explore incorporating external influencing factors and adaptive mechanisms to further enhance the model's forecasting accuracy and generalization capabilities.

CRediT authorship contribution statement

Yang Yang: Writing – review & editing, Data curation. **Yuchao Gao:** Writing – review & editing, Formal analysis. **Hu Zhou:** Writing – original draft, Visualization, Methodology. **Jinran Wu:** Writing – review & editing, Supervision, Methodology, Formal analysis, Data curation, Conceptualization. **Shangce Gao:** Writing – review & editing. **You-Gan Wang:** Writing – review & editing, Supervision.

Data and code availability

The datasets used in this study will be made available upon reasonable request to the corresponding author.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Performance comparison across different datasets for long-term forecasting

In Section 4, experiments are conducted on four groups of power load datasets from Portugal, Australia, America, and New England to evaluate the MG-Autoformer. Additionally, the model is applied to six groups of time series datasets for further comparison.

The results in Table A.8 demonstrate that MG-Autoformer outperforms the standard Autoformer and other baseline models across the majority of datasets and forecasting horizons. Specifically, MG-Autoformer achieves superior prediction accuracy and enhanced stability on the Electricity, Exchange, Weather, and ETT datasets. These datasets exhibit clear periodic patterns and relatively stable long-term trends, which align well with the MG-ACAM and its ability to capture cross-scale temporal dependencies.

Notably, MG-Autoformer also demonstrates improved performance on the ILI dataset, particularly for shorter-horizon forecasts such as the 24-step and 36-step predictions. These improvements can be attributed to the model's ability to capture seasonal patterns and partially periodic characteristics. However, on the Traffic dataset, MG-Autoformer's performance is less competitive, reflected in higher MSE and MAE than Autoformer. This can be explained by the Traffic dataset's more irregular and less predictable temporal patterns, which lack clear periodicity. Such characteristics inherently weaken the effectiveness of MG-Autoformer's multi-granularity mechanisms, highlighting that model performance can vary significantly across datasets with different temporal structures, noise levels, and degrees of regularity.

Overall, the results confirm that MG-Autoformer provides substantial performance advantages for long-term deterministic forecasting when applied to datasets with well-defined seasonal and multi-scale temporal structures. At the same time, the performance on Traffic highlights that MG-Autoformer is not universally superior, reinforcing the importance of model selection based on dataset characteristics rather than assuming the existence of a universal best model.

Data availability

Data will be made available on request.

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